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Optimal redundancy allocation to maximize multi-state computer network reliability subject to correlated failures

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ABSTRACT

Modern society depends on the stability of computer networks. One way to achieve this goal is to determine the optimal redundancy allocation such that system reliability is maximized. Redundancy requires that each edge in computer networks possess several binary-state physical lines allocated in parallel. A computer network implementing redundancy allocation is called a multi-state computer network (MSCN), since each edge can exhibit multiple states with a probability distribution according to the number of binary-state physical lines that are operational. However, past research often fails to consider the possibility of correlated failures. This study applies a correlated binomial distribution to characterize the state distribution of each edge within a network and a redundancy optimization approach integrating simulated annealing (SA), minimal paths, and correlated binomial distribution is proposed. The approach is applied to four practical computer networks to demonstrate the computational efficiency of the proposed SA relative to several popular soft computing algorithms.

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1. Introduction

In modern society, computer networks are the main medium for data transmission. Accidents or repair may disrupt network operation, preventing the execution of normal operations. Preserving network stability is thus important to ensure uninterrupted service. Therefore, many system supervisors focus on system reliability assessment and maximization for the computer networks they operate. The issues underlying system reliability assessment can be traced to binary-state computer network reliability assessment. In a binary-state computer network, each edge denoting a transmission line, may either work or not. System reliability is defined as the probability that the source vertex is connected to the sink vertex [6,16]. Reliability assessment can also be extended to multi-state computer networks (MSCNs), where each edge is composed of several physical lines and each physical line may provide a fixed capacity (i.e. bandwidth) or fail. Clearly, such MSCNs are more detailed than their binary state counterparts. In the multi-state systems, reliability is defined as the probability that a computer network can successfully transmit d units of data from source to sink [17,18,34].

Since system reliability is regarded as a performance index to evaluate the capability of a computer network, most studies discuss system reliability maximization for MSCNs from the perspective of network topology optimization [5]; [28] or component assignment optimization [19,21]. Network topology optimization focuses on exploring the connections between edges and vertices to search for the optimal network topology that maximizes system reliability. According to Stevenson [30], topology optimization is often expensive since it involves designing or redesigning networks. Component assignment optimization considers a set of multi-state components available for allocation to edges with the goal of determining the optimal component assignment that maximizes system reliability. The advantage of component assignment optimization is that system reliability can be improved without altering the existing network topology. However, in such situations, each component's maximal redundancy is fixed (i.e. each component's redundancy level is given) such that an edge may provide surplus capacity after allocating a component.

Redundancy allocation is another way to maximize system reliability for computer networks. The concept of redundancy allocation was introduced by Misra and Sharma [24]. It requires that each edge consists of several binary-state components allocated in parallel to enhance system stability. In other words, redundancy allocation determines the redundancy level for each edge and has been applied in many fields. Huang et al. [12] adopted a redundancy technique to promote the reliability for an aircraft's multi-channel electrical power supply system. Tian et al. [31] presented an optimization model for a multi-state series-parallel

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Nomenclature

E; n	set of edges connecting pair of vertices; number of edges
V	set of vertices
(**E**, **V**)	computer network
O, D	unique source vertex; unique sink vertex
e_i	edge #i for $i=1, 2, \ldots, n$
k	capacity of a physical line
p	successful probability of physical line
c	cost per unit of length of physical line
y_i	redundancy level of edge e_i, $i=1, 2, \ldots, n$
$\boldsymbol{Y}$	$(y_1, y_2, \ldots, y_n)$: a redundancy allocation or solution
$C_{\boldsymbol{Y}}$	allocation cost associated with $\boldsymbol{Y}$
$G_{i,w}$	state #w of edge e_i, $w=1, 2, \ldots, y_i+1$, where $G_{i,1}=0$
$\hat{G}_i$	maximum capacity of edge e_i
x_i	current capacity of e_i, $i=1, 2, \ldots, n$
$\boldsymbol{X}$	$(x_1, x_2, \ldots, x_n)$: (current) capacity vector
$\boldsymbol{M_Y}$	$(\hat{G}_1, \hat{G}_2, \ldots, \hat{G}_n)$: maximal capacity vector associated with $\boldsymbol{Y}$
$\mathbf{U_Y}$	set of $\boldsymbol{X}$ feasible associated with $\boldsymbol{Y}$
d	demand at the sink vertex D
$\mathbf{X_Y}$	set of $\boldsymbol{X}$ fulfilling d
$R_d(\boldsymbol{Y})$	system reliability given component allocation $\boldsymbol{Y}$
B	allocation budget
k	transmission capacity per unit of redundancy level
ρ_i	correlation among physical lines along edge e_i
γ	performance level of an edge
β_i	notational simplification
m	number of MPs
P_j	MP #j, $j=1, 2, \ldots, m$
f_j	amount of flow through P_j, $j=1, 2, \ldots, m$
$\boldsymbol{F}$	$(f_1, f_2, \ldots, f_m)$: data flow pattern
$\mathbf{F_{MY}}$	set of F feasible under $\boldsymbol{M_Y}$
$\mathbf{F_X}$	set of F feasible under X with $X \in \mathbf{U_Y}$
τ	iteration #τ
θ	number of neighbors
PR	probability to accept the best neighborhood solution
T_0	initial temperature
T_τ	control temperature in iteration #τ
α	cooling rate
Ψ	termination time

system to jointly determine the optimal component state distribution and optimal redundancy for each stage. Azaron et al. [3] solved a multi-objective discrete reliability optimization problem in a cold-standby redundant system using a genetic algorithm (GA). Tian et al. [32] developed a practical approach for the joint reliability–redundancy optimization of multi-state series-parallel systems. The approach not only determines the optimal redundancy level for each parallel subsystem, but also finds the optimal values for the variables that affect the component state distributions in each subsystem. Liu et al. [22] presented an approach of joint redundancy and imperfect maintenance strategy optimization for multi-state systems. Feizollahi et al. [8] proposed a robust optimization framework to deal with uncertain component reliabilities in redundancy allocation problems for series-parallel systems. Roy et al. [29] discussed a multi-objective reliability redundancy allocation problem for the series-parallel system and found the optimum number of redundant components to maximize system reliability and minimize system cost with entropy as the constraint. These studies demonstrated that an advantage of redundancy allocation is that each edge's redundancy level is appropriately determined so that the edge has no surplus capacity as much as possible.

In this study, redundancy allocation of an edge determines the number of physical lines allocated to the edge, i.e. redundancy level. For example, an edge with redundancy level 2 combines 2 physical lines. Since each physical line possesses two states, the edge has three states (i.e. no physical line works, one of physical line works, and both physical lines work) following a binomial distribution. Most studies [17-19,21] assume the physical lines allocated to an edge are statistically independent. However, failure of a physical line may cause other lines in the edge to partially or completely fail. For example, a disaster may cause multiple physical lines to fail, necessitating repair. During repair of a physical line, other lines belonging to the edge need to be partially or fully suspended. Such a phenomenon is called a correlated failure and should not be ignored in system reliability assessment and optimization [9,10].

This paper determines the optimal redundancy allocation of a computer network to maximize system reliability subject to a budget, while considering the impact of correlated physical-line failures. To our knowledge, this problem has not been considered previously. Chambari et al. [4] illustrated that redundancy allocation optimization is a NP-hard problem. Thus, the problem considered here is also NP-hard. In order to solve the optimal redundancy allocation problem with correlated failures, a simulated annealing (SA) approach is proposed based on the conceptual approach of Chambari et al. [4]. SA was introduced by Metropolis et al. [23] and has been widely applied to combinatorial optimization problems [33]. SA stimulates the annealing process of solids to find a near optimal solution and has the advantage of avoiding convergence to local optimum. According to the strategy suggested by Dong et al. [7], we integrate 1-opt and 2-opt operations in the stage of neighborhood search of the proposed SA to enhance its ability to perform global search. Moreover, a correlated binomial distribution model developed by Fiondella and Zeephongsekul [10] is utilized to evaluate the probability distribution for each edge under redundancy allocation when failures are correlated, and the system reliability is evaluated in terms of minimal paths (MP). The purposes of this study are to propose and solve the system reliability redundancy allocation problem with correlated failures and to demonstrate the computational efficiency of the proposed SA by comparing it with GA, particle swarm optimization (PSO), and tabu search (TS) through case studies of four practical computer networks.

The remainder of the paper is organized as follows. Assumptions and problem formulation are detailed in Section 2. State distribution of a correlated physical line redundancy allocation is illustrated in Section 3. Multi-state computer network model associated with a redundancy allocation is constructed in Section 4. Section 5 describes the proposed SA. Numerical experiments of four practical computer networks are executed to compare the proposed SA with several popular soft computing algorithms in Section 6. The conclusions are given in Section 7, along with future research.

2. Assumptions and problem formulation

Let $\mathbf{E}=\{e_i | 1 \le i \le \text{le}; n\}$ be a set of edges, where e_i denotes the ith edge and **V** be a set of vertices. Thus, a computer network is represented as (**E**, **V**). The length of e_i is denoted by l_i for $i=1, 2, \ldots, n$. Moreover, let k be the capacity of a single physical line. The

probability that a physical line can successfully transmit data (i.e., it can provide the capacity k) is denoted by p. The cost to transmit per unit of length of a physical line is denoted by c. A *redundancy allocation* is represented as a vector $\boldsymbol{Y}=(y_1, y_2, \ldots, y_n)$ where y_i is the redundancy level in edge e_i for $i=1, 2, \ldots, n$. For instance, the term "$y_1=2$" implies that edge e_1 is composed of two physical lines. Thus, edge e_1 can exhibit one of three states: (i) both physical lines are operational, (ii) one physical line is operational and the other failed, and (iii) both physical lines are failed. Therefore, the computer network $(\mathbf{E}, \mathbf{V})$ associated with the allocation $\boldsymbol{Y}$ can be characterized as an MSCN.

The cost per unit of length of a physical line is denoted by c. Thus, the cost to allocate one physical line to edge e_i is measured in terms of its length l_i, i.e., $c \times l_i$. Given the redundancy level y_i, the aggregate cost of all of the physical lines allocated to edge e_i is $c \times l_i \times y_i$. Therefore, the allocation cost can be expressed as

$$C_{\boldsymbol{Y}} = \sum_{i=1}^{n} \left(c \times l_i \times y_i\right). \tag{1}$$

Definition 1. $\boldsymbol{X} \leq \boldsymbol{U}$ implies $(x_1, x_2, \ldots, x_n) \leq (u_1, u_2, \ldots, u_n)$. That is, $x_i \leq u_i$ for each i.

Definition 2. $\boldsymbol{X} < \boldsymbol{U}$ implies $(x_1, x_2, \ldots, x_n) < (u_1, u_2, \ldots, u_n)$ which satisfies $\boldsymbol{X} \leq \boldsymbol{U}$ and $x_i < u_i$ for at least one i.

A redundancy level of y_i in edge e_i corresponds to y_i+1 states with capacities, $0=G_{i,1} < G_{i,2} < \ldots < G_{i,yi+1}=\hat{G}_i$ and the probability of an edge being in one of these y_i+1 states can be characterized with an arbitrary discrete probability distribution. The capacity of edge e_i in a given state is $G_{i,w}=k(w-1)$ for $w=1, 2, \ldots, y_i+1$ and $\hat{G}_i=G_{i,yi+1}$. For convenience, let $\boldsymbol{M_Y}=(\hat{G}_1, \hat{G}_2, \ldots, \hat{G}_n)$ be the maximal capacity vector associated with $\boldsymbol{Y}$. A capacity vector denoted by $\boldsymbol{X}=(x_1, x_2, \ldots, x_n)$ indicates the current state of computer network $(\mathbf{E}, \mathbf{V})$, where x_i represents the current capacity of edge e_i for $i=1, 2, \ldots, n$. The capacity vector $\boldsymbol{X}$ is not deterministic because x_i is a random variable taking the values $G_{i,1}, G_{i,2}, \ldots$, and $\hat{G}_i$ for $i=1, 2, \ldots, n$. Any capacity vector $\boldsymbol{X}$ must satisfy the following constraint:

$$\boldsymbol{X} \leq \boldsymbol{M_Y}. \tag{2}$$

Here, the constraint in Eq. (2) indicates that the current capacity x_i cannot exceed the maximal capacity $\hat{G}_i$ for each e_i. Let $\mathbf{U_Y}$ be the set of $\boldsymbol{X}$ satisfying constraint (2) associated with redundancy allocation $\boldsymbol{Y}$ and $V(\boldsymbol{X})$ be the maximal flow of $(\mathbf{E}, \mathbf{V})$ under $\boldsymbol{X}$, where the maximal flow is defined as the largest amount of data that the computer network $(\mathbf{E}, \mathbf{V})$ can transmit from O to D given $\boldsymbol{X}$. Suppose d is the amount of data the source O must send to the sink D. System reliability associated with redundancy allocation $\boldsymbol{Y}$ is defined as the probability that the maximal flow of the network is greater than or equal to d and is denoted $R_d(\boldsymbol{Y})=\Pr\{\boldsymbol{X}|V(\boldsymbol{X}) \geq d, \boldsymbol{X} \in \mathbf{U_Y}\}$. For notational convenience, the set $\{\boldsymbol{X}|V(\boldsymbol{X}) \geq d, \boldsymbol{X} \in \mathbf{U_Y}\}$ is designated as $\boldsymbol{X_Y}$ and thus $R_d(\boldsymbol{Y})=\Pr\{\boldsymbol{X}|\ \boldsymbol{X} \in \boldsymbol{X_Y}\}$.

The mathematical programming formulation of the optimal redundancy allocation problem with maximal reliability for an MSCN is therefore:

$$\textit{Maximize } R_d(\boldsymbol{Y}) = \Pr\{\boldsymbol{X}|\boldsymbol{X} \in \mathbf{X}_Y\} \tag{3}$$

Subject to

$$y_i \geq 0 \quad \text{for } i = 1,\ 2,\ \ldots,\ n,\ \text{and} \tag{4}$$

$$C_{\boldsymbol{Y}} \leq B, \tag{5}$$

where B denotes a budget.

The constraint in Eq. (4) states each redundancy should be larger than or equal to zero. If the optimal redundancy allocation takes $y_\omega=0$, edge e_ω can be removed from the network topology. The constraint in Eq. (5) indicates that the cost of each redundancy allocation $\boldsymbol{Y}$ should be smaller than or equal to budget B. Thus, Eqs. (4) and (5) explicitly define the feasible redundancy allocations. The optimal redundancy allocation $\boldsymbol{Y}$ is the one that maximizes system reliability $R_d(\boldsymbol{Y})$, which can be obtained by maximizing the objective function given in Eq. (3). To solve the optimization problem, several assumptions are given as follows:

I. Flow in $(\mathbf{E}, \mathbf{V})$ must satisfy the flow-conservation law [11], which means that no flow will reduce or increase during the transmission.
II. Each physical line has the same capacity, success probability, and allocation (deployment) cost.
III. The states of different edges are statistically independent for any redundancy allocation.

3. State distribution of correlated physical line redundancy allocation

Before developing an MSCN model associated with a redundancy allocation $\boldsymbol{Y}$, we describe an edge state distribution where the physical lines may experience correlated failure. Consider an edge e_i with redundancy level y_i and each physical line providing capacity k in the operational state. The reliability and unreliability of each physical line is characterized by the probabilities p and $(1-p)$ respectively. When the failures of the physical lines are uncorrelated, the probability distribution of edge e_i can be represented by the binomial distribution.

$$\Pr\{x_i = k\gamma = G_{i,\gamma+1}\} = \binom{y_i}{\gamma}(p)^{\gamma}(1-p)^{y_i-\gamma}, \tag{6}$$

where γ denotes a performance level which takes one of the integer values: 0, 1, …, y_i. The term, $k\gamma$, denotes the capacity provided by γ physical lines. Eq. (6) indicates the probability that the current capacity is equal to the γ operational physical lines.

Correlated failure may cause the physical lines allocated to the same edge to fail in a statistically dependent manner. Thus, Eq. (6) may not accurately characterize the failure behavior in all situations. Let ρ_i represent the correlation among the physical lines along edge e_i. Generally, the closer the edge e_i is to the location where an event or disaster occurs, the higher correlation ρ_i will be. According to the correlated binomial distribution proposed by Fiondella and Zeephongsekul [10], the probability distribution of edge e_i can be represented as

$$\Pr\{x_i = k\gamma = G_{i,\gamma+1}\} = \frac{1}{\beta_i}\binom{y_i}{\gamma}\left(p\beta_i\right)^{\gamma}(1-p\beta_i)^{y_i-\gamma}. \tag{7}$$

where $\beta_i=(1+\rho_i(1-p)/p)$. Eq. (7) indicates the probability the current capacity is equal to γ physical lines' possessing correlation ρ_i. Moreover, the probability the current capacity is greater than or equal to γ physical lines possessing correlation ρ_i is given by Eq. (8).

$$\Pr\{x_i \geq k\gamma = G_{i,\gamma+1}\} = \frac{1}{\beta_i}\sum_{v=\gamma}^{y_i}\binom{y_i}{v}(p\beta_i)^{v}(1-p\beta_i)^{y_i-v}. \tag{8}$$

The probability that the current capacity x_i is 0, i.e., $\gamma=0$ is

$$\Pr\{x_i \geq 0 = G_{i,1}\} = 1-\frac{1}{\beta_i}\sum_{v=1}^{y_i}\binom{y_i}{v}(p\beta_i)^{v}(1-p\beta_i)^{y_i-v}, \tag{9}$$

where the term, $\frac{1}{\beta_i}\sum_{v=1}^{y_i}\binom{y_i}{v}(p\beta_i)^{v}(1-p\beta_i)^{y_i-v}$, is the probability that

Table 1
Probability distributions of edge e_i with four physical lines and different correlations.

Capacity of edge e_i	Correlation value ρ_i					
	0	0.2	0.4	0.6	0.8	1
0 Gbps (i.e., $\gamma=0$)	0.00160	0.04824	0.09110	0.13047	0.16667	0.2
4 Gbps (i.e., $\gamma=1$)	0.02560	0.01311	0.00553	0.00164	0.00020	0
8 Gbps (i.e., $\gamma=2$)	0.15360	0.10322	0.06083	0.02826	0.00737	0
12 Gbps (i.e., $\gamma=3$)	0.40960	0.36127	0.29737	0.21668	0.11796	0
16 Gbps (i.e., $\gamma=4$)	0.40960	0.47416	0.54518	0.62295	0.70779	0.8

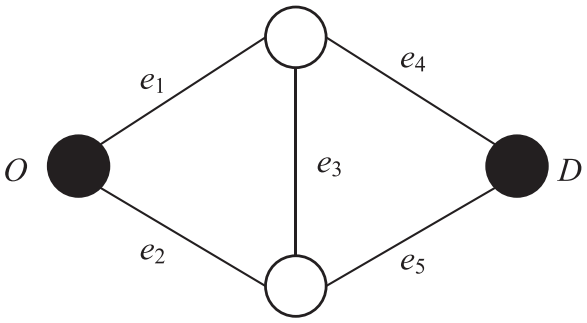


Fig. 1. A simple network.

Table 2
Probability distribution.

Edge e_i	Capacity		
	0 Gbps	1 Gbps	2 Gbps
e_1	0.028	0.144	0.828
e_2	0.100	0.900	0
e_3	0.055	0.090	0.855
e_4	0.019	0.162	0.819
e_5	0.100	0.900	0

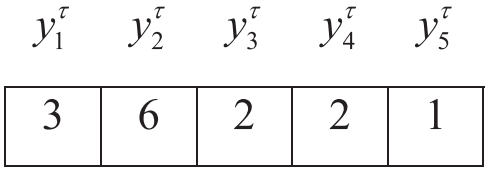


Fig. 2. Solution representation.

at least one physical line on edge e_i works. The probability distribution of each edge associated with a redundancy allocation $\boldsymbol{Y}$ can be determined via Eqs. (7)–(9).

Suppose that each available physical line provides a capacity of 4 Gbps (Gigabit per second) with probability $p=0.8$ and there are four physical lines allocated to edge e_i. Table 1 provides the probability distributions of the edge e_i for different correlations using Eqs. (7)–(9). In practice, correlations can be inferred from sources of hazard specific historical data such as records of natural disasters like earthquakes or hurricanes. Moreover, the analytical nature of the expressions given here enable "what if" analysis to consider the impact correlation would have on a given network design. Thus, optimization studies utilizing correlations can identify designs that would be robust despite correlated failures and therefore serve as an effective planning aid.

4. Multi-state computer network model associated with a redundancy allocation

This section describes the relationship between the flow and capacity associated with a redundancy allocation to build an MSCN model. We then propose an algorithm to measure system reliability.

4.1. Relationship between capacity and data flow

Suppose there are a total of m MPs in $(\mathbf{E}, \mathbf{V})$: $P_1, P_2, \ldots, P_m$. A data flow pattern $\boldsymbol{F}=(f_1, f_2, \ldots, f_m)$ describes the data transmission through the MSCN, where f_j is the flow passing via P_j. The following constraint characterizes the fact that the total flow cannot exceed the maximal capacity of e_i associated with redundancy allocation $\boldsymbol{Y}$, where $\sum_{j:e_i\in P_j} f_j$ is the total data flow passing through edge e_i. In other words, any $\boldsymbol{F}$ satisfying Constraint (10) is equivalent to the $\boldsymbol{F}$ feasible under $\boldsymbol{M_Y}$.

$$\sum\nolimits_{j:e_i\in P_j} f_j \le \hat{G}_{y_n} \text{ for each } i = 1,\ 2,\ \ldots,\ n, \tag{10}$$

For convenience, let $\mathbf{F}_{\boldsymbol{MY}}$ denote the set of $\boldsymbol{F}$ feasible under $\boldsymbol{M_Y}$. Similarly, let $\mathbf{F}_{\boldsymbol{X}}$ denote the set of $\boldsymbol{F}$ feasible under $\boldsymbol{X}$ with $\boldsymbol{X}\in\mathbf{U}_{\boldsymbol{Y}}$, i.e., the set of F fulfilling Constraint (11).

$$\sum\nolimits_{j:e_i\in P_j} f_j \le \boldsymbol{X} \text{ for each } i = 1,\ 2,\ \ldots,\ n, \tag{11}$$

Therefore, the maximal data flow of $(\mathbf{E}, \mathbf{V})$ under $\boldsymbol{X}$ may be expressed as $V(\boldsymbol{X})=\max\{\sum_{j=1}^{m} f_j | \boldsymbol{F}\in\mathbf{F}_{\boldsymbol{X}}\}$.

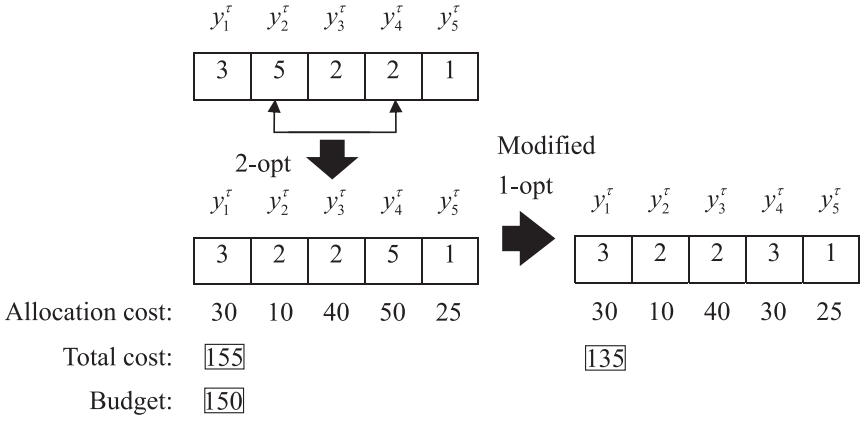


Fig. 3. Neighborhood search.

4.2. System reliability measurement

The system reliability associated with $\boldsymbol{Y}$ is denoted by $R_d(\boldsymbol{Y})=\{\boldsymbol{X} \mid V(\boldsymbol{X}) \geq d, \boldsymbol{X}\in\mathbf{U}_{\boldsymbol{Y}}\}=\Pr\{\boldsymbol{X} \mid \boldsymbol{X}\in\mathbf{X}_{\boldsymbol{Y}}\}$. However, enumerating all $X\in\mathbf{X}_{\boldsymbol{Y}}$ and then summing their probabilities is not efficient. In order to avoiding searching for $\boldsymbol{X}\in\mathbf{X}_{\boldsymbol{Y}}$, the smallest capacity vector to transmit d units of data, named d-SV, is defined below and is employed to increase the computational efficiency of system reliability [18].

Definition 3. A d-SV $\boldsymbol{X}$ is the smallest element in $\mathbf{X}_{\boldsymbol{Y}}$ such that $\boldsymbol{U}\notin\mathbf{X}_{\boldsymbol{Y}}$ for any capacity vector $\boldsymbol{U}$ with $\boldsymbol{U}<d$-SV $\boldsymbol{X}$.

Suppose there are q d-SV $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_q$ in total. That is, the set $\{\boldsymbol{X}|\boldsymbol{X}\in\mathbf{X}_{\boldsymbol{Y}}\}$ can be replaced by the union of sets in terms of d-SV, i.e., $\{\boldsymbol{X}|\boldsymbol{X}\in\mathbf{X}_{\boldsymbol{Y}}\} = \{\cup_{i=1}^{q}\{\mathrm{X}|\mathrm{X}\geq \mathrm{X}_i, \mathrm{X}\in\mathbf{X}_{\boldsymbol{Y}}\}\}$, in which case system reliability can be expressed as $R_d(\boldsymbol{Y})=\Pr\{\cup_{i=1}^{q}\{\mathrm{X}|\mathrm{X}\geq \mathrm{X}_i, \mathrm{X}\in\mathbf{X}_{\boldsymbol{Y}}\}\}$. In order to calculate the probability, the recursive sum of disjoint products (RSDP) is employed since it exhibits greater computational efficiency than the inclusion-exclusion method and state space decomposition technique [37].

The following algorithm is developed based on the correlated binomial distribution, d-SV, and RSDP to evaluate system reliability associated with $\boldsymbol{Y}$.

Reliability calculation algorithm (RCA). //calculate system reliability for $R_d(\boldsymbol{Y})$.

Step 1. Calculate each edge's probability distribution associated with $\boldsymbol{Y}$ according to the correlated binomial distribution model.
Step 2. Find all $\boldsymbol{F}$ satisfying demand d (Constraint (12))and maximal capacity vector $\boldsymbol{M}_{\boldsymbol{Y}}$ (Constraint (13)). If no feasible F exists, then $R_d(\boldsymbol{Y})=0$ and quit the algorithm.

$$\sum_{j=1}^{m} f_j = d. \tag{12}$$

$$\sum_{j:e_i\in P_j} f_j \leq \hat{G}_{\gamma n} \quad \text{for each } i = 1,\ 2,\ \ldots,\ n, \tag{13}$$

Step 3. Transform each $\boldsymbol{F}$ into $\boldsymbol{X}$ via the following Eq. (14).

$$x_i = G_{i,w} \text{ where } G_{i,w} \geq \sum_{j:e_i\in P_j} f_j > G_{i,w-1},$$
$$w \in \{1,\ 2,\ \ldots,\ y_i + 1\} \text{ for } i = 1,$$
$$2,\ \ldots,\ n. \tag{14}$$

Step 4. Compare all capacity vectors obtained from Step 3 and then preserve the minimal ones, i.e., all d-SV.
Step 5. Suppose $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_q$ are q d-SV from Step 4. Use the RSDP approach to calculate $R_d(\boldsymbol{Y})=\Pr\{\cup_{i=1}^{q}\{\mathrm{X}|\mathrm{X}\geq \mathrm{X}_i, \mathrm{X}\in\mathbf{X}_{\boldsymbol{Y}}\}\}$.

4.3. A simple example to illustrate the procedure to calculate system reliability

Consider a simple network with 5 edges and 4 MPs: $P_1=\{e_1, e_4\}$, $P_2=\{e_1, e_3, e_5\}$, $P_3=\{e_2, e_3, e_4\}$, and $P_4=\{e_2, e_5\}$ (see Fig. 1). Suppose that each physical line provides 1 Gbps with $p=0.9$ or may fail. Given $\boldsymbol{Y}=(2, 1, 2, 2, 1)$, $d=2$, and $\boldsymbol{\rho}=(\rho_1, \rho_2, \rho_3, \rho_4, \rho_5)=(0.2, 0.2, 0.5, 0.1, 0.1)$, the RCA to calculate reliability is illustrated step by step as below:

Step 1. Under $\boldsymbol{Y}=(2, 1, 2, 2, 1)$ and $\boldsymbol{\rho}=(0.2, 0.2, 0.5, 0.1, 0.1)$, each edge's probability distribution associated with $\boldsymbol{Y}$ is determined in Table 2 according to the correlated binomial distribution model.
Step 2. Determine all $\boldsymbol{F}$ satisfying $d=2$ and maximal capacity vector $\boldsymbol{M}_{\boldsymbol{Y}}=(2, 1, 2, 2, 1)$.

$$f_1 + f_2 + f_3 + f_4 = 2 \text{ and} \tag{15}$$
$$f_1 + f_2 \leq 2$$
$$f_3 + f_4 \geq 1$$
$$f_2 + f_3 \leq 2$$
$$f_1 + f_3 \leq 2$$
$$f_2 + f_4 \leq 1. \tag{16}$$

Thus, there are five feasible data flow vectors: $\boldsymbol{F}_1=(1, 0, 0, 1)$, $\boldsymbol{F}_2=(1, 0, 1, 0)$, $\boldsymbol{F}_3=(1, 1, 0, 0)$, $\boldsymbol{F}_4=(2, 0, 0, 0)$, and $\boldsymbol{F}_5=(0, 1, 1, 0)$.
Step 3. Transform each feasible $\boldsymbol{F}$ into $\boldsymbol{X}$ via the following equation.

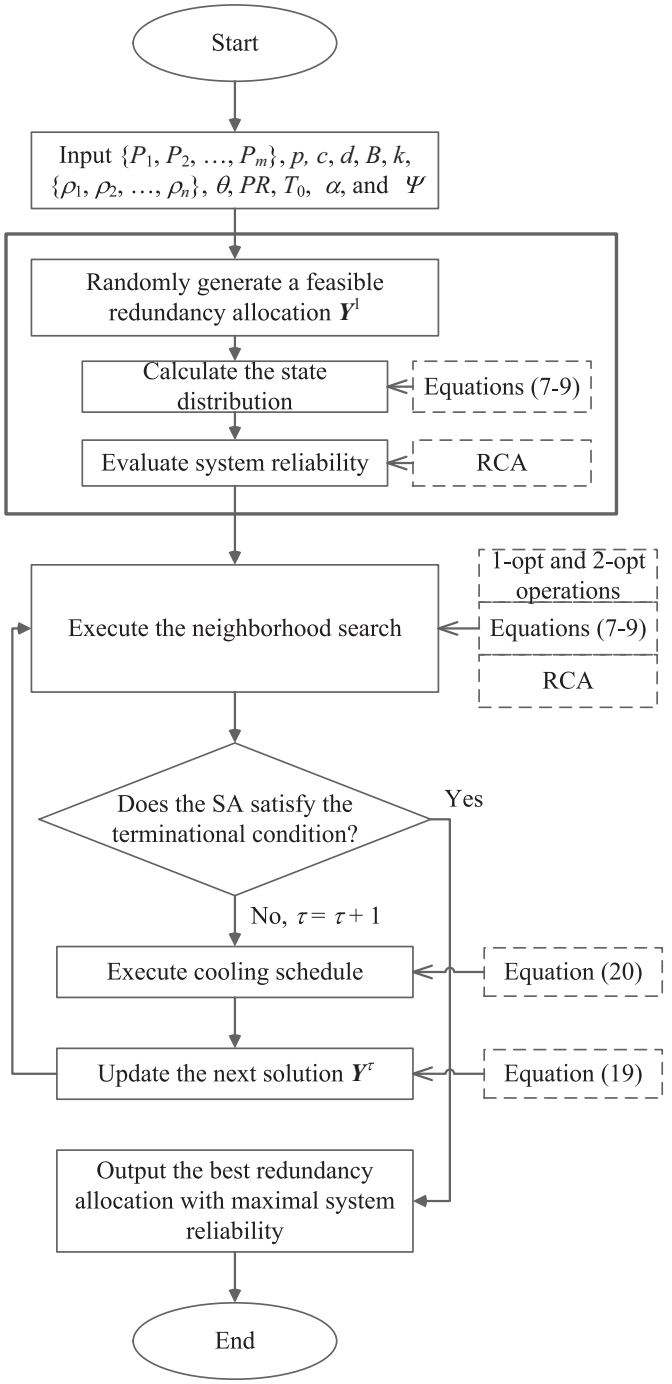


Fig. 4. The proposed SA procedure.

$x_1 = G_{1,w}$ where $G_{1,w} \geq f_1 + f_2 > G_{1,w-1}$, $w \in \{1, 2, 3\}$,
$x_2 = G_{2,w}$ where $G_{2,w} \geq f_3 + f_4 > G_{2,w-1}$, $w \in \{1, 2\}$,
$x_3 = G_{3,w}$ where $G_{3,w} \geq f_2 + f_3 > G_{3,w-1}$, $w \in \{1, 2, 3\}$,
$x_4 = G_{4,w}$ where $G_{4,w} \geq f_1 + f_3 > G_{4,w-1}$, $w \in \{1, 2, 3\}$,
$x_5 = G_{5,w}$ where $G_{5,w} \geq f_2 + f_4 > G_{5,w-1}$, $w \in \{1, 2\}$. (17)

Then, we obtain five capacity vectors: $\boldsymbol{X}_1=(1, 1, 0, 1, 1)$, $\boldsymbol{X}_2=(1, 1, 1, 2, 0)$, $\boldsymbol{X}_3=(2, 0, 1, 1, 1)$, $\boldsymbol{X}_4=(2, 0, 0, 2, 0)$, $\boldsymbol{X}_5=(1, 1, 2, 1, 1)$.
Step 4. Compare these five capacity vectors. Since $\boldsymbol{X}_5$ is larger than $\boldsymbol{X}_1$, it is not a 2-SV. In addition, no element in the set $\{\boldsymbol{X}_1, \boldsymbol{X}_2, \boldsymbol{X}_3, \boldsymbol{X}_4, \boldsymbol{X}_5\}$ is smaller than $\boldsymbol{X}_1$ (resp. $\boldsymbol{X}_2$, $\boldsymbol{X}_3$,$\boldsymbol{X}_4$), and therefore it is a 2-SV.
Step 5. There are four 2-SV: $\boldsymbol{X}_1=(1, 1, 0, 1, 1)$, $\boldsymbol{X}_2=(1, 1, 1, 2, 0)$, $\boldsymbol{X}_3=(2, 0, 1, 1, 1)$, and $\boldsymbol{X}_4=(2, 0, 0, 2, 0)$. System reliability $R_2(\boldsymbol{Y})=\Pr\{\cup_{i=1}^{4}\{\mathrm{X}|\mathrm{X} \geq \mathrm{X}_i, \mathrm{X} \in \boldsymbol{\mathrm{X}}_{\boldsymbol{Y}}\}\}$ calculated by the RSDP is 0.92264.

5. Simulated annealing

The concept of SA is adopted from the "annealing" process used in the metallurgical industry. Initially, a solid is heated, and then a slow cooling schedule is executed to allow crystallization to a minimal energy. During the initial execution of the SA, the temperature should be high enough such that the probability of dropping in the local optimum is low. As the execution of cooling schedule progresses, the SA gradually converges to a near-optimum solution. To implement the proposed SA to solve the optimal redundancy allocation for multi-state network reliability despite correlated failures, solution representation, neighborhood search, cooling schedule, and terminal condition are discussed in the following subsections.

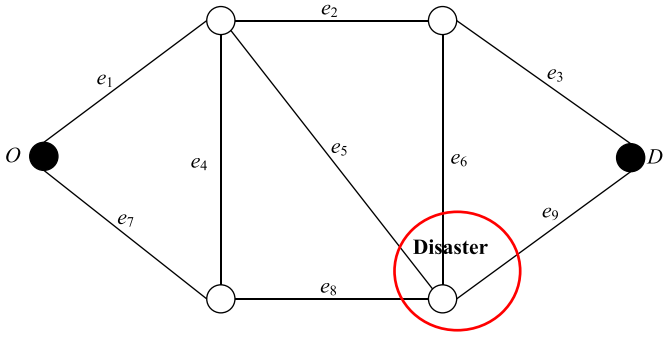


Fig. 5. ARPA topology [27].

5.1. Solution representation

The constraints of the problem should be considered in solution representation as far as possible. Since a solution to the addressed problem represent a redundancy allocation, the notation $\boldsymbol{Y}$ continues to be adopted in the proposed SA but is modified to be $\boldsymbol{Y}^{\tau}=(y_1^{\tau}, y_2^{\tau}, \ldots, y_n^{\tau})$ by adding a superscript τ, where $\boldsymbol{Y}^{\tau}$ is the current solution at the τth iteration. Each solution includes n variables and is encoded by integer values. For instance, Fig. 2 descripts a solution for the network with 5 edges. The term, $y_1^{10}=3$, signifies that there are 3 physical lines allocated to edge e_1 at the 10th iteration.

Initially, the proposed SA generates a solution $\boldsymbol{Y}^1$ randomly in which each variable y_i^1 should satisfy constraint (4), i.e. $y_i^1 \geq 0$. If the initial solution's allocation cost exceeds budget B, the variable with the maximal value will be decreased by 1. For instance, if the allocation cost of initial solution $\boldsymbol{Y}^1=(3, 6, 2, 2, 1)$ exceeds the budget, y_2^1 is chosen and then decreased to 5. Thus, $\boldsymbol{Y}^1=(3, 5, 2, 2, 1)$. This continues until the budget constraint is satisfied. Moreover, in order to promote the generation of a feasible solution, we adopt a constraint that each variable y_i^{τ} should satisfy demand d in the solution encoding.

$$k\, y_i^{\tau} \leq d \text{ for } i = 1, 2, \ldots, n. \quad (18)$$

For instance, if $k=1$ and $d=4$, then $y_i^{\tau}=5$ should be adjusted to be $y_i^{\tau}=4$. Similarly, any current solution $\boldsymbol{Y}^{\tau}$ should be checked for feasibility according to the above encoding principle.

5.2. Neighborhood search, cooling schedule, and terminal condition

The most commonly-used neighborhood searches are 1-opt and 2-opt operations [7]. Both operations are simple local search techniques. The 1-opt operation randomly selects an edge e_i and then changes its redundancy level y_i^{τ} to be another value. The 2-opt operation randomly selects two edges e_i and e_j and then swaps their redundancy levels. Dong et al. [7] illustrated that the 1-opt and 2-opt operations possess the abilities to diversify and intensify the search, respectively. Therefore, this study integrates both operations and then proposes a neighborhood search (see Fig. 3). At first, the 2-opt operation is executed to generate a neighborhood solution. Subsequently, if the neighborhood solution does not satisfy the budget constraint, the modified 1-opt operation is utilized to select the edge whose allocation cost is greater than the others and then replaces its redundancy level

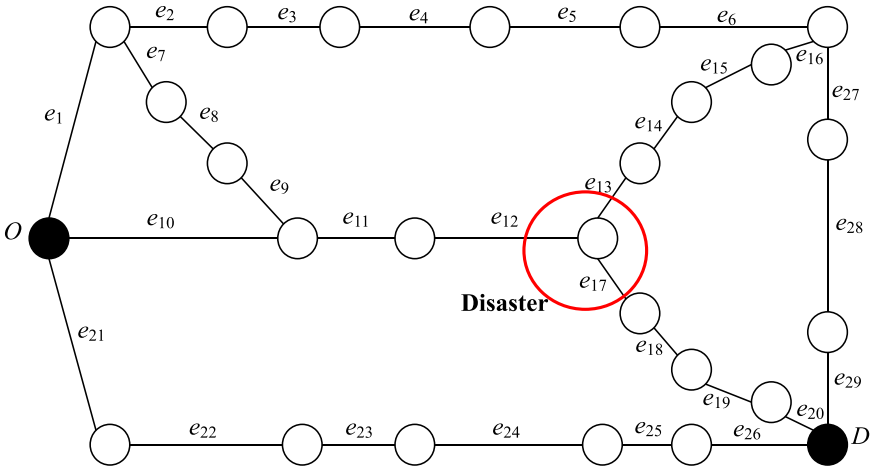


Fig. 6. OCT topology [2].

with a smaller random value. Moreover, the modified 1-opt operation is executed until the neighborhood solution satisfies the budget.

The neighborhood search generates a set of neighborhood solutions. While using the 2-opt operation, the larger number of neighbors is C_2^n. However, it is time-consuming to enumerate C_2^n neighbors as n increases. Referring to [4,25], an integer parameter θ denoting the number of repetitions allowed at each temperature level is defined to control the size of the neighborhood of solutions found by the proposed neighborhood search. This parameter is similar to the number of neighbors in the typical TS [15,26], and should not exceed C_2^n.

In the set of neighborhood solutions, if the best neighbor's system reliability is greater than the other neighbors but less than the current solution's system reliability, then this neighboring solution may be accepted with acceptance probability PR. This probability is calculated with the following commonly-used equation [14].

$$PR = \exp\left(\frac{-\Delta R}{T_\tau}\right), \tag{19}$$

where ΔR represents the difference between the best neighbor's system reliability and the current maximal system reliability. The notation T_τ denotes the current control temperature which is calculated by the following equation.

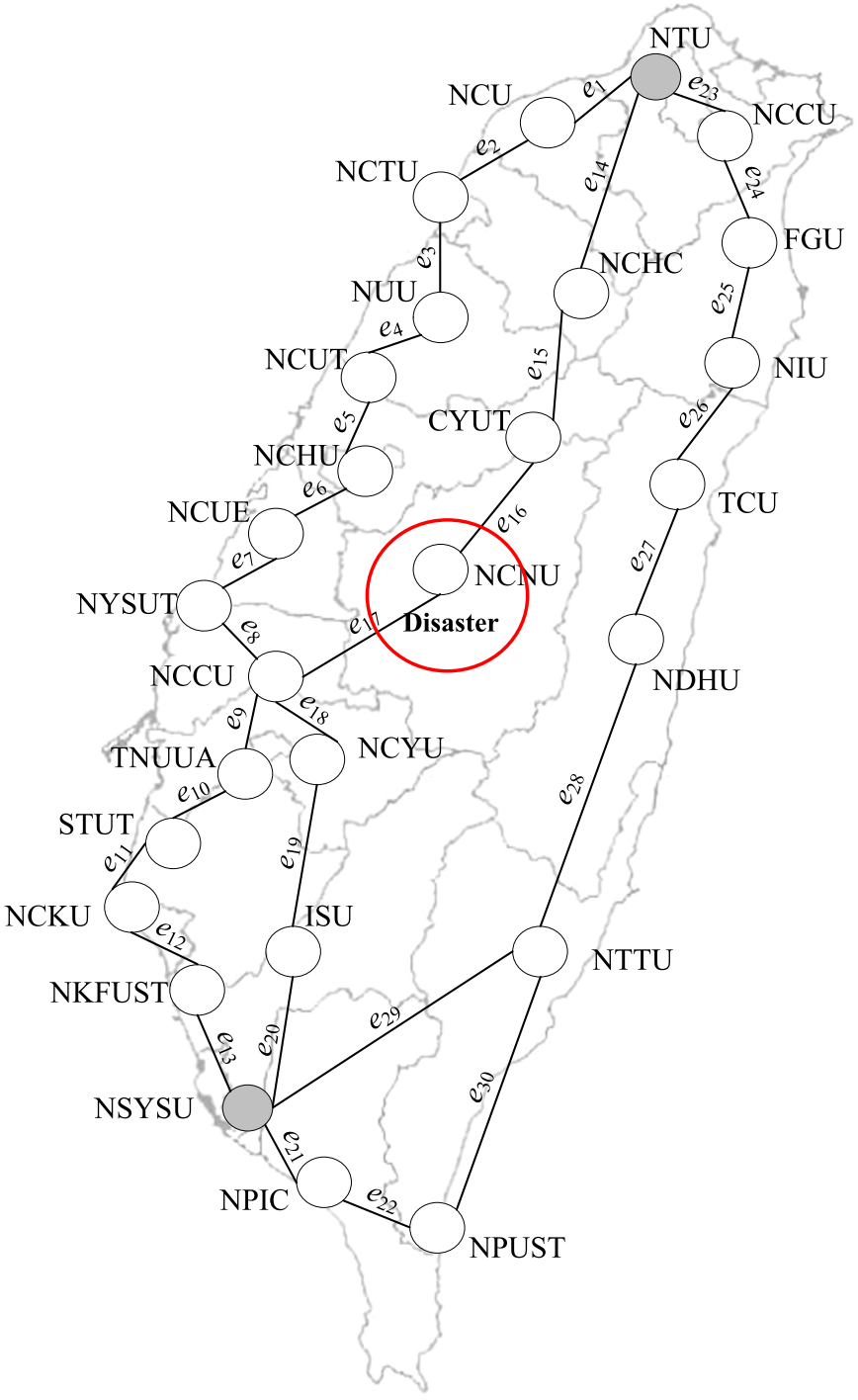


Fig. 7. TANET [36].

$$T_t = \alpha T_{t-1}, \tag{20}$$

where the notation α denotes the cooling rate. As τ increases, the control temperature decreases. Such a process is called the cooling schedule. Subsequently, a random value in the interval [0,1] is generated and compared with the probability *PR*. If *PR* is greater than or equal to the random value, the current solution is replaced by the best neighbor as the next solution $\boldsymbol{Y}^{\tau+1}$.

Since most system supervisors expect to obtain a near-optimum solution in a reasonable period, this paper considers the execution time as the termination condition. That is, the proposed SA continues until the CPU time exceeds to the termination time.

5.3. The proposed SA procedure

Based on the illustration of the SA proposed in Sections 5.1 and 5.2, its procedure is as shown in Fig. 4 and described as follows.

Input: $\{P_1, P_2, \ldots, P_m\}$ (set of MPs), p (successful probability of a physical line), c (cost per unit of length of a physical line), d (demand), B (allocation budget), k (capacity of a physical line), $(\rho_1, \rho_2, \ldots, \rho_n)$ (vector of correlations), θ (number of neighbors), *PR* (probability to accept the best neighborhood solution), T_0 (initial temperature), α (cooling rate), and Ψ (termination time).

Output: Maximum system reliability and best redundancy allocation

Step 1. Initially, $\tau = 1$. Randomly generate a feasible redundancy allocation $\boldsymbol{Y}^1 = (y_1^1, y_2^1, \ldots, y_n^1)$.
Step 2. Calculate the state distribution of $\boldsymbol{Y}^1$ under $\{\rho_1, \rho_2, \ldots, \rho_n\}$ via Eqs. (7)–(9).
Step 3. Evaluate system reliability of $\boldsymbol{Y}^1$ using RCA and let $\boldsymbol{Y}^1$ be the best current solution.
Step 4. Execute the neighborhood search.
41. Utilize the modified operation of hybrid 1-opt and 2-opt to search θ neighbors of the current solution $\boldsymbol{Y}^\tau$.
42. For all neighborhood solutions, calculate their state distributions under $\{\rho_1, \rho_2, \ldots, \rho_n\}$ via Eqs. (7)–(9) and then evaluate their system reliabilities using RCA.
43. Find the best neighborhood solution whose system reliability is larger than the other neighbors.
44. If the best neighborhood solution is better than the current best solution, replace the best current solution.

Step 5. If the CPU time is larger than or equal to Ψ, output the best current redundancy allocation with maximal system reliability and stop the proposed SA; otherwise, continue the SA and $\tau = \tau + 1$.
Step 6. Execute cooling schedule to obtain $T_\tau = \alpha T_{\tau-1}$.
Step 7. Update the next solution $\boldsymbol{Y}^\tau$ for executing the neighborhood search.
71. If the best neighborhood solution is better than the current best solution, let the best neighborhood solution be the next solution $\boldsymbol{Y}^\tau$ for executing the neighborhood search; otherwise, execute the following Steps 7.2–7.3.
72. Calculate the acceptance probability $PR = \exp\left(\frac{-\Delta R}{T_\tau}\right)$.
73. Generate a random number $\sigma \sim U(0, 1)$. If $\sigma \le PR$, let the best neighborhood solution be the next solution $\boldsymbol{Y}^\tau$. Go to Step 4.

6. Numerical experiments

In this section, the proposed SA is applied to four practical computer networks which are ARPA, OCT, TANET, and NSFNET. Figs. 5–8 depict their network topologies. ARPA is composed of 9 edges, 4 vertices, and 13 MPs, OCT is composed of 29 edges, 24 vertices, and 9 MPs, TANET is composed of 30 edges, 27 vertices,

Table 3
Length and correlation data of ARPA.

(a) Length of each link in Fig. 5

edge: e_i	Length (km): l_i	edge: e_i	Length (km): l_i	edge: e_i	Length (km): l_i
e_1	30	e_4	25	e_7	29
e_2	16	e_5	42	e_8	16
e_3	29	e_6	25	e_9	30

(b) Correlation of each edge in Fig. 5

edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i
e_1	0.10	e_4	0.23	e_7	0.15
e_2	0.37	e_5	0.65	e_8	0.67
e_3	0.35	e_6	0.73	e_9	0.70

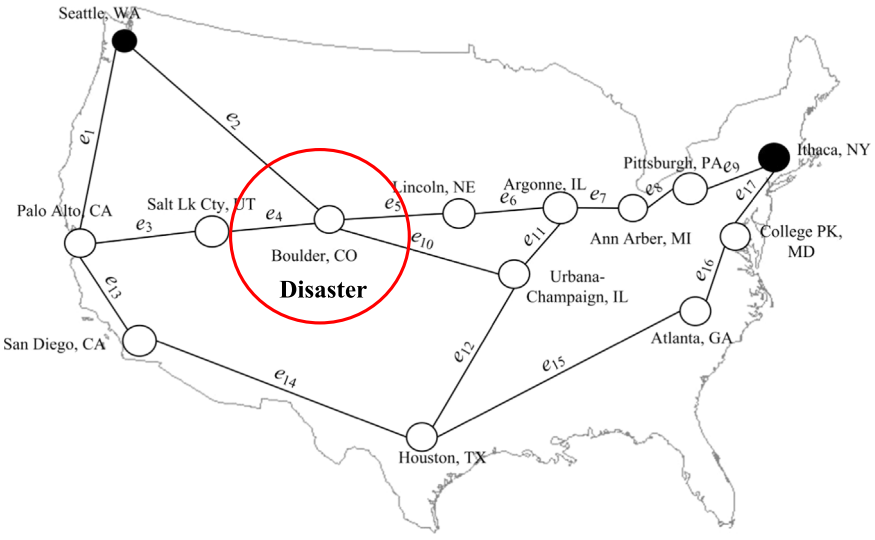


Fig. 8. NSFNET [20]. (a) Analysis for initial temperature (T_0). (b) Analysis for number of neighbors (θ). (c) Analysis for cooling rate (α).

Table 4
Length and correlation data of OCT.

(a) Length of each edge in Fig. 6.

edge: e_i	Length (km): l_i	edge: e_i	Length (km): l_i	edge: e_i	Length (km): l_i
e_1	29	e_{11}	20	e_{21}	30
e_2	9	e_{12}	25	e_{22}	27
e_3	10	e_{13}	16	e_{23}	14
e_4	11	e_{14}	17	e_{24}	26
e_5	12	e_{15}	17	e_{25}	8
e_6	12	e_{16}	16	e_{26}	17
e_7	15	e_{17}	16	e_{27}	20
e_8	13	e_{18}	18	e_{28}	31
e_9	14	e_{19}	18	e_{29}	24
e_{10}	27	e_{20}	15		

(b) Correlation of each edge in Fig. 6.

edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i
e_1	0	e_{11}	0.41	e_{21}	0
e_2	0	e_{12}	0.63	e_{22}	0
e_3	0.09	e_{13}	0.68	e_{23}	0.11
e_4	0.13	e_{14}	0.59	e_{24}	0.16
e_5	0.22	e_{15}	0.53	e_{25}	0.32
e_6	0.38	e_{16}	0.39	e_{26}	0.47
e_7	0	e_{17}	0.70	e_{27}	0.39
e_8	0	e_{18}	0.54	e_{28}	0.42
e_9	0.17	e_{19}	0.49	e_{29}	0.41
e_{10}	0.10	e_{20}	0.45		

Table 5
Length and correlation data of TANET.

(a) Length of each link in Fig. 7

edge: e_i	Length (km): l_i	edge: e_i	Length (km): l_i	edge: e_i	Length (km): l_i
e_1	40	e_{11}	21	e_{21}	17
e_2	49	e_{12}	35	e_{22}	52
e_3	60	e_{13}	17	e_{23}	37
e_4	49	e_{14}	92	e_{24}	53
e_5	32	e_{15}	93	e_{25}	49
e_6	33	e_{16}	64	e_{26}	114
e_7	44	e_{17}	73	e_{27}	101
e_8	36	e_{18}	28	e_{28}	95
e_9	58	e_{19}	76	e_{29}	33
e_{10}	20	e_{20}	25	e_{30}	52

(b) Correlation of each edge in Fig. 7

edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i
e_1	0.04	e_{11}	0.36	e_{21}	0.09
e_2	0.09	e_{12}	0.25	e_{22}	0.14
e_3	0.18	e_{13}	0.21	e_{23}	0.02
e_4	0.34	e_{14}	0.31	e_{24}	0.08
e_5	0.63	e_{15}	0.59	e_{25}	0.16
e_6	0.72	e_{16}	0.86	e_{26}	0.19
e_7	0.74	e_{17}	0.93	e_{27}	0.33
e_8	0.75	e_{18}	0.62	e_{28}	0.37
e_9	0.61	e_{19}	0.47	e_{29}	0.40
e_{10}	0.52	e_{20}	0.22	e_{30}	0.15

and 6 MPs, and NSFNET is composed of 16 edges, 14 vertices, and 6 MPs. The length data of the four networks are given in Tables 3–6 (a), respectively. ARPA and OCT are cited from [2,27], respectively. TANET is an academic computer network connecting all Taiwanese educational or academic organizations (e.g. elementary schools, high schools, universities, and research centers), where each vertex represents an organization [36]. This study treats NTU (resp. NSYSU) as the source (resp. sink) vertex. NSFNET, built by the National Center for Supercomputing Applications at the University of Illinois and Cornell Theory Center, is a network backbone [20]. This study focuses on the data transmission from Seattle to Ithaca. Hypothetical disaster prone areas are marked with a red circle in Fig. 5 (resp. Figs. 6–8). Generally, edges close to this area may possess greater correlation. Edge correlations are given in Tables 3–6(b).

Table 6
Length and correlation data of NSFNET.

(a) Length of each link in Fig. 8

edge: e_i	Length (km): l_i	link: a_i	Length (km): l_i
e_1	997	e_{10}	1477
e_2	1546	e_{11}	219
e_3	1121	e_{12}	1293
e_4	824	e_{13}	877
e_5	782	e_{14}	1981
e_6	793	e_{15}	1682
e_7	630	e_{16}	1186
e_8	781	e_{17}	515
e_9	483		

(b) Correlation of each edge in Fig. 8

edge: e_i	Correlation: ρ_i	edge: e_i	Correlation: ρ_i
e_1	0.45	e_{10}	0.66
e_2	0.33	e_{11}	0.43
e_3	0.75	e_{12}	0.36
e_4	0.78	e_{13}	0.09
e_5	0.63	e_{14}	0.12
e_6	0.22	e_{15}	0
e_7	0.06	e_{16}	0
e_8	0	e_{17}	0
e_9	0		

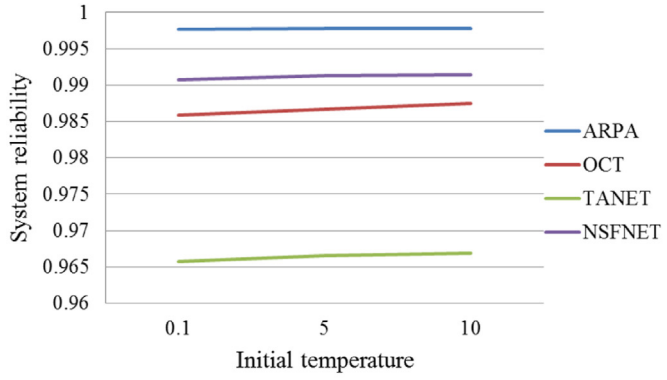
Table 7
SA parameter analysis for computer networks considered.

No.	Parameter setting (T_0, α, θ)	**Average** maximal system reliability			
		Fig. 5 (ARPA)	Fig. 6 (OCT)	Fig. 7 (TANET)	Fig. 8 (NSFNET)
01	(0.1, 0.925, 10)	0.99762	0.98432	0.96455	0.99168
02	(0.1, 0.925, 20)	0.99767	0.99125	0.96986	0.99114
03	(0.1, 0.925, 30)	0.99740	0.98556	0.96435	0.99153
04	(0.1, 0.95, 10)	0.99747	0.98626	0.96480	0.99031
05	(0.1, 0.95, 20)	0.99784	0.98539	0.96311	0.98925
06	(0.1, 0.95, 30)	0.99787	0.98432	0.96574	0.99042
07	(0.1, 0.975, 10)	0.99789	0.98346	0.97044	0.99063
08	(0.1, 0.975, 20)	0.99775	0.98663	0.96393	0.99064
09	(0.1, 0.975, 30)	0.99748	0.98536	0.96505	0.99083
10	(5, 0.925, 10)	0.99761	0.98611	0.96704	0.99093
11	(5, 0.925, 20)	0.99741	0.98711	0.96648	0.99266
12	(5, 0.925, 30)	0.99773	0.98535	0.96770	0.99136
13	(5, 0.95, 10)	0.99737	0.98895	0.96723	0.99096
14	(5, 0.95, 20)	0.99766	0.98437	0.96774	0.99001
15	(5, 0.95, 30)	0.99800	0.98661	0.96588	0.99226
16	(5, 0.975, 10)	0.99773	0.98736	0.96338	0.99216
17	(5, 0.975, 20)	0.99802	0.98758	0.96809	0.99209
18	(5, 0.975, 30)	0.99807	0.98641	0.96542	0.98923
19	(10, 0.925, 10)	0.99753	0.98638	0.96484	0.99231
20	(10, 0.925, 20)	**0.99817**	0.98385	0.96509	0.99233
21	(10, 0.925, 30)	0.99796	0.98811	0.96586	0.98911
22	(10, 0.95, 10)	0.99781	0.98626	0.96691	0.99067
23	(10, 0.95, 20)	0.99811	0.98696	**0.97352**	0.99154
24	(10, 0.95, 30)	0.99787	0.98661	0.96658	**0.99273**
25	(10, 0.975, 10)	0.99732	0.99054	0.96599	0.99107
26	(10, 0.975, 20)	0.99748	0.98537	0.96670	0.99139
27	(10, 0.975, 30)	0.99773	**0.99293**	0.96666	0.99166

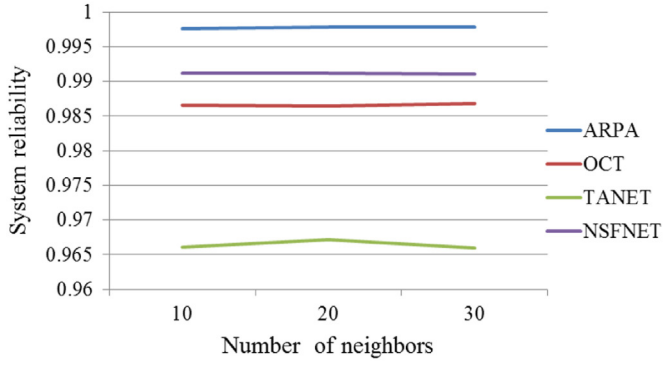
*Bold value indicates the largest average maximal system reliability identified.

There are 80 OC-18 (Optical Carrier 18) lines to be allocated. Each OC-18 line owns the capacity of approximate 1 Gbps (giga bits per second) as it operates with probability 0.986 and 0 Gbps with probability 0.014 according to the database provided by a telecommunication service provider. An OC-18 line cost 30 NTD (new Taiwan dollars) per unit of length. Using the four practical computer networks and 80 OC-18 lines, several numerical

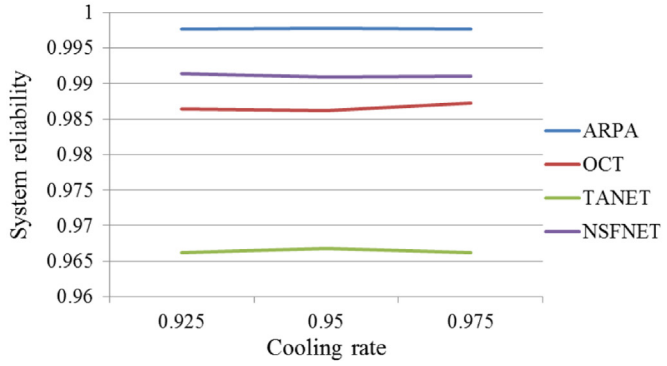
experiments are implemented to analyze the parameters of the proposed SA, and to compare it with several soft-computing algorithms including GA, PSO, and TS. All algorithms were programmed in the MATLAB programming language and executed on a personal computer with 64-bit Windows 10, Intel Core i7-4712MQ CPU 2.3 GHz, and 8 GB RAM.



(a) Analysis for initial temperature (T_0).



(b) Analysis for number of neighbors (θ).



(c) Analysis for cooling rate (α).

Fig. 9. Analysis for different SA parameters.

6.1. Analysis of SA parameters

The SA parameter settings influence the computational efficiency. Several important SA parameters including number of neighbors (θ), initial temperature (T_0), and cooling rate (α) are considered for each of the four practical computer networks.

For each network, we consider 5 Gb data (i.e., $d=5$) to be transferred from the source vertex to the sink vertex. Due to different topologies, the budgets given to ARPA, OCT, TANET, and NSFNET are 17,000 NTD, 42,000, 125,000 NTD, and 1,200,000 NTD, respectively. Referring to [15,26], the number of neighbors (θ) considered is in the range [5,20]. The initial temperature (T_0) should be high enough in the first iteration such that the probability of accepting a worse solution is at least 0.8 [1,14]. Most studies, such as [1,4,7], suggested that the cooling rate (α) should be smaller than 1 and is usually in the interval [0.9, 1). Based on the recommended parameter settings, we consider 27 combinations of $\theta\in\{10, 15, 20\}$, $T_0\in\{0.1, 5, 10\}$, and $\alpha\in\{0.925, 0.95, 0.975\}$, and fix the termination time with $\Psi=3600$ s. For each combination, the proposed SA is implemented for 20 trials and then the average maximal system reliability is calculated. The experimental results are given in Table 7. While focusing on the changes of initial temperature (T_0) without considering the changes of number of neighbors and cooling rate, it is found that the average maximal system reliability increases as T_0 increases for the four practical networks (see Fig. 9). While focusing on the changes in the number of neighbors (θ) (resp. cooling rate (α)) without considering the other SA parameters, the average maximal system reliability is not changed significantly as it increases for the four networks. Actually, the 27 parameter combinations perform well when determining the maximal system reliability since the difference between the largest average maximal system reliability and the smallest average maximal system reliability does not exceeds 0.0105 for TANET (resp. 0.01 for ARPA, OCT, and NSFNET). In other words, these parameter combinations coincide with the suggestions from the previous studies and are usable for solving the optimal redundancy allocation problem with the proposed SA. For each network, the best parameter combination is also highlighted in Table 7. These combinations are also used in the following subsection.

6.2. Computational efficiency analysis for different soft computing algorithms

In order to demonstrate computational efficiency, this subsection compares the proposed SA with GA, PSO, and TS on the ARPA, OCT, TANET, and NSFNET, where the GA is developed based on [19], PSO is developed based on [35], TS is developed based on [13]. The obtained best SA parameter settings continue to be adopted. Under $d=5$ and $B=17{,}000$ (resp. $B=42{,}000$, $B=125{,}000$, and $B=1{,}200{,}000$), each algorithm is executed for 20 trials to obtain the largest maximal system reliability, average maximal system reliability, and average iteration number for ARPA (resp. OCT, TANET, and NSFNET), where each trial was given 3600 s to find the best solution.

Table 8
Experimental results of Fig. 5 via four soft computing algorithms.

Soft computing algorithm	Average maximal reliability	Average iteration number	Best redundancy allocation	Corresponding cost	Largest maximal reliability
GA	0.98463	1068	(5, 5, 5, 1, 1, 1, 1, 2, 1)	16,740	0.98463
PSO	0.97705	1923	(5, 5, 5, 1, 1, 1, 1, 2, 1)	16,740	0.98463
SA	**0.99848***	2142	(3, 2, 2, 2, 3, 2, 2, 2, 2)	16,680	**0.99897**
TS	0.96999	2192	(5, 0, 1, 1, 4, 1, 1, 0, 1)	16,710	0.98214

* The bold value represents the largest one while comparing with the other algorithms.

Table 9
Experimental results of Fig. 6 via four soft computing algorithms.

Soft computing algorithm	Average maximal reliability	Average iteration number	Best redundancy allocation	Corresponding cost	Largest maximal reliability
GA	0.98607	8826	(3, 3, 3, 3, 3, 3, 1, 2, 1, 3, 3, 3, 1, 1, 1, 3, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3)	41,820	0.99189
PSO	0.98492	6634	(2, 3, 4, 3, 3, 3, 1, 2, 1, 3, 3, 3, 2, 2, 1, 1, 3, 3, 3, 3, 4, 4, 4, 3, 4, 3, 2, 2, 2)	41,880	0.99151
SA	**0.99158**	8213	(2, 2, 3, 3, 3, 2, 3, 1, 2, 3, 3, 3, 2, 1, 2, 1, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3)	41,760	**0.99318**
TS	0.97660	8350	(3, 4, 1, 1, 2, 4, 2, 4, 1, 1, 0, 3, 3, 3, 3, 2, 3, 3, 3, 3, 4, 4, 4, 4, 2, 1, 2, 3, 2)	40,200	0.98579

Table 10
Experimental results of Fig. 7 via four soft computing algorithms.

Soft computing algorithm	Average maximal reliability	Average iteration number	Best redundancy allocation	Corresponding cost	Largest maximal reliability
GA	0.95698	31,336	(1, 1, 1, 1, 1, 1, 1, 2, 3, 4, 4, 4, 4, 5, 5, 4, 4, 4, 4, 5, 2, 1, 4, 2, 3, 2, 2, 2, 2, 1)	124,920	0.95775
PSO	0.94638	24,485	(3, 2, 2, 3, 2, 3, 2, 2, 3, 3, 4, 4, 4, 3, 3, 3, 3, 4, 4, 4, 2, 2, 3, 3, 3, 2, 2, 2, 2, 2)	124,830	0.95206
SA	**0.96721**	32,691	(4, 1, 2, 1, 3, 1, 1, 1, 1, 1, 1, 3, 3, 3, 4, 2, 2, 3, 2, 2, 4, 2, 4, 4, 4, 4, 4, 3, 3, 1)	122,220	**0.96759**
TS	0.92503	24,738	(0, 1, 1, 5, 1, 2, 0, 2, 0, 1, 1, 4, 3, 2, 2, 0, 3, 5, 1, 1, 1, 2, 4, 3, 1, 2, 4, 3, 3, 2)	124,410	0.94163

Table 11
Experimental results of Fig. 8 via four soft computing algorithms.

Soft computing algorithm	Average maximal reliability	Average iteration number	Best redundancy allocation	Corresponding cost	Largest maximal reliability
GA	0.95597	1186	(3, 3, 1, 1, 3, 3, 4, 4, 4, 1, 3, 1, 2, 2, 3, 2, 3)	1,199,430	0.95705
PSO	0.95619	2766	(1, 5, 2, 1, 3, 3, 4, 5, 5, 2, 3, 1, 1, 1, 2, 2, 2)	1,196,580	0.96059
SA	**0.99215**	2771	(2, 3, 2, 3, 3, 3, 2, 3, 2, 2, 3, 2, 2, 2, 2, 2, 3)	1,195,020	**0.99320**
TS	0.81097	3418	(3, 3, 3, 2, 3, 2, 2, 2, 2, 3, 2, 3, 3, 2, 0, 3, 0)	1,194,300	0.84348

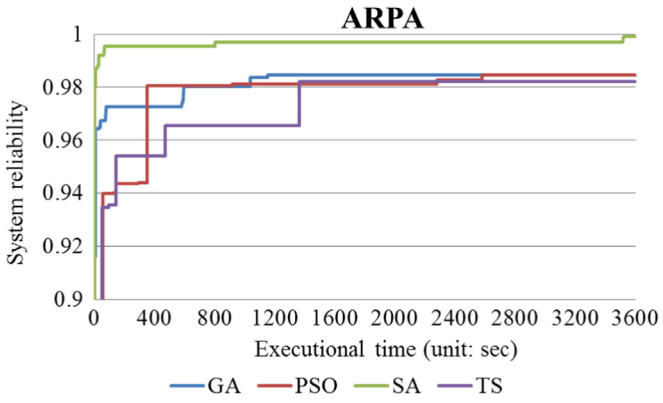


Fig. 10. Comparison of GA, PSO, SA, and TS for Fig. 5.

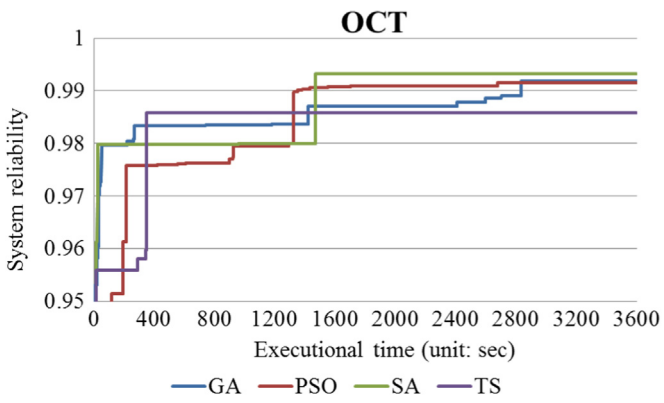


Fig. 11. Comparison of GA, PSO, SA, and TS for Fig. 6.

The experimental results are summarized in Tables 8–11. However, there is no guarantee that algorithms executed for additional iterations would not have discover better solutions. The proposed SA clearly achieves the best computational performance when solving the redundancy allocation problem. In addition, observing the best solutions from the TS, there exists several redundancy levels with 0, e.g. $\boldsymbol{Y}=(5, 0, 1, 1, 4, 1, 1, 0, 1)$ for ARPA. Such a solution simplifies the network's topology but lowers system reliability. Figs. 10–13 depict the solution-search process for each algorithm. The proposed SA finds near optimal solutions faster than the other approaches. However, it occasionally converges to local optimum. For instance, in TANET (resp. NSFNET), the solution obtained by the SA does not significantly improve after 853 (resp. 1283) seconds, but this may be explained by the fact that it

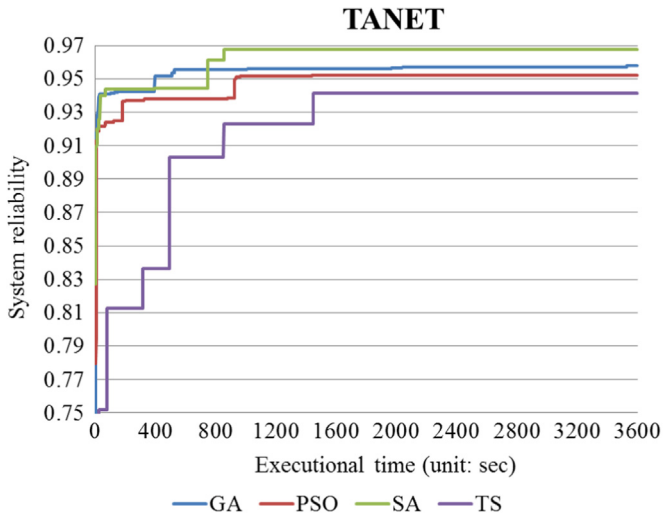


Fig. 12. Comparison of GA, PSO, SA, and TS for Fig. 7.

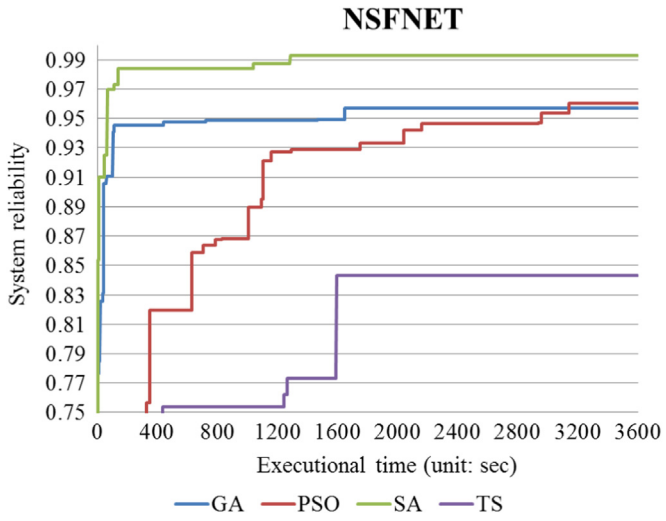


Fig. 13. Comparison of GA, PSO, SA, and TS for Fig. 8.

converges to a superior solution more quickly. The TS performance is the poorest and therefore not suggested to solve the redundancy allocation problem considered here.

7. Conclusions

This study discusses a physical line redundancy allocation optimization problem to maximize system reliability subject to a budget, where the physical lines allocated to the same edge may experience correlated failure. Considering failure correlation is more realistic than the majority of previous studies and the redundancy allocation strategy to maximize system reliability is more convenient than topology redesign and is also more flexible than the component allocation. The proposed SA integrating MP and correlated binomial distribution has been validated to exhibit better computational efficiency than GA, PSO, and TS. The best redundancy allocation obtained can be regarded as a strategy to improve the transmission stability of a computer network. Moreover, the redundancy allocation strategy not only maximizes system reliability but can also simplify network topology.

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